MATH 1110
Calculus 1: Quiz 3
Given Time: 2 hrs.

Name: _____ Student ID: _____

Instruction. For each of the following problems, show all necessary steps in your work. Answers submitted without supporting work or with illegible handwriting may not be graded. Each question is worth the number of points indicated. You have **90 minutes** to complete the exam.

1. [5 points] Let $f(x) = x^2 - x - 12$. Use the definition of the derivative to show that $f'(x) = 2x - 1$.

2. [5 points] Find an equation of the tangent line to the graph of $f(x) = x^3 + 1$ at $x = 1$.

3. [20 points] Find the derivative of the following functions.

a) $f(x) = x^3 - 3x^2$

b) $g(x) = (x^3 + 4x^2 - 1)(x^2 - 3x + 1)$

c) $h(x) = \frac{x^4+1}{2x^2+3x-1}$

d) $F(x) = \ln(\sinh x)$

e) $G(x) = \cos\left(\tan^{-1} x\right)$

4. [15 points] Find the second derivative of the following functions.

a) $f(x) = x^2 e^x$

b) $g(x) = 2x^3 - 3x^2 + \frac{2}{x^2}$

c) $h(x) = \cosh(4x) - \sin(3x)$

5. [10 points] a) Show that if $f(x)$ is differentiable at $x = c$, then $f(x)$ is continuous at $x = c$.

a) If $f(x)$ is continuous at $x = c$, is $f(x)$ differentiable at $x = c$? If yes, prove it. If no, give .a counterexample.

6. [5 points] Use implicit differentiation to find the slope of the tangent line to the $x^3 + 2xy + y^3 = 4$ at $(1, 1)$.

7. [10 points] The length of a rectangle is increasing at a rate of $8cm/s$ and its width is increasing at a rate of $4cm/s$. When the length is $20cm$ and the width is $10cm$, how fast is the area of the rectangle increasing?

8. [10 points] If a rock is thrown upward on the planet Mars with a velocity of $10m/s$, its height (in meters) after t seconds is given by $H(t) = 10t - 1.86t^2$.

a) Find the velocity of the rock after one second.

b) Find the velocity of the rock when $t = a$

c) When will the rock hit the surface?

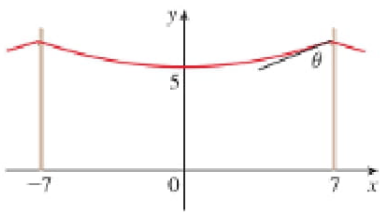
d) With what velocity will the rock hit the surface?

9. [10 points] One ship is approaching a port from the north at 10 miles per hour, and is presently 3 miles north of the port. A second ship has already left the port, traveling east at 15 miles per hour, and is presently 4 miles east of the port. At this instant, what is the rate of change of the distance between the two ships? Are they getting closer or further apart?

10. [10 points] A telephone line hangs between two poles $14m$ apart in the shape of the catenary $y = 20\cosh(x/20) - 15$, where x and y are measured in meters.

a) Find the slope of this curve where it meets the right-hand pole.

b) Find the angle θ between the line and the pole.



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